Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 7 Tutorial Worksheet 1: Firewall Fundamentals, Types and Evolution

Field	Detail
Based on	Week 7 Deck A, Firewall Fundamentals lecture (definitions, IDS/IPS contrast, software vs hardware, history, stateless vs stateful, seven firewall types, evolution)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate and compare security technologies)
Format	12 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's Firewall Fundamentals lecture. Several ask you to compare or recommend between options rather than simply describe one, which is deliberate: LO4 is specifically about evaluating and comparing security technologies, not just naming them. Write full answers, not one-line definitions.

Foundational Concepts

Q1. Explain the difference between a firewall, an IDS, and an IPS, specifically in terms of what each one actually does once it identifies something suspicious. *[LO4]*

Q2. Explain the difference between a software firewall and a hardware firewall, and why most real networks deploy both together rather than choosing one over the other. *[LO4]*

Q3. Explain the difference between stateless and stateful inspection, and describe one specific scenario where a stateless firewall's lack of context becomes a genuine vulnerability. *[LO4]*

The Seven Firewall Types

Q4. Distinguish Packet Filtering from Stateful Inspection specifically in terms of how each treats return traffic for a connection that was originally initiated from inside the network. *[LO4]*

Q5. Explain what a Circuit-Level Gateway actually validates, and why it provides no protection against malicious content once a session has already been approved. *[LO3]*

Q6. A company wants deep content filtering, such as blocking specific content inside an HTTP request, and strong anonymisation of its internal users' identities. Evaluate whether an Application-Layer/Proxy firewall or a simple Stateful Inspection firewall is the better fit, and justify your answer. *[LO4]*

Q7. Distinguish NGFW from UTM. Both bundle multiple security functions together, so what is the key architectural difference between them, and what trade-off does that difference actually create? *[LO4]*

Q8. A small business wants comprehensive security, firewall, antivirus, and content filtering, but has no dedicated security staff and a limited budget. Recommend either an NGFW or a UTM for this business, and justify your recommendation using each option's genuine trade-offs. *[LO3, LO4]*

Q9. An e-commerce company already has a stateful inspection firewall at its perimeter but has recently suffered a SQL injection attack against its public web application. Evaluate why the existing firewall failed to stop this, and recommend a specific additional firewall type that would address it. *[LO3, LO4]*

Bringing It Together

Q10. Explain why NGFW emerged specifically once port-based rules stopped meaningfully reflecting real traffic. What actually changed about real-world network traffic that made this true? *[LO4]*

Q11. WAF and UTM both emerged as responses to increasing complexity and more specialised threats, but they represent opposite strategies. Explain what these two strategies are, and give an example of an organisation that would be better served by each one. *[LO4]*

Q12. Using the full arc of firewall evolution covered this week, from packet filtering through to NGFW, explain what is being traded for what at every step. What consistently goes down, and what consistently goes up in return for it? *[LO4]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. Several questions above ask you to recommend one option over another for a specific scenario. Practise stating your recommendation first, then justifying it, since that is exactly the structure a structured-essay exam question rewards.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5). The evaluation-style questions above are close in format to the structured-essay section.